

Table 1: Initial electric-field magnitude comparison after the OPALX one-step BeamBeam field gather.

| Particle | Species  | Pair | $S$ [mm] | $ E _{\text{OPALX}}$ [V/m] | $ E _{\text{manuf.}}$ [V/m] | Rel. err.                |
|----------|----------|------|----------|----------------------------|-----------------------------|--------------------------|
| Te-1     | electron | 1    | -0.9000  | $1.497157 \times 10^7$     | $2.686889 \times 10^7$      | $-4.4279 \times 10^{-1}$ |
| Te+1     | positron | 1    | -0.9000  | $1.497157 \times 10^7$     | $2.686889 \times 10^7$      | $-4.4279 \times 10^{-1}$ |
| Te-2     | electron | 2    | -0.5994  | $3.221653 \times 10^8$     | $3.288006 \times 10^8$      | $-2.0180 \times 10^{-2}$ |
| Te+2     | positron | 2    | -0.5994  | $3.221653 \times 10^8$     | $3.288006 \times 10^8$      | $-2.0180 \times 10^{-2}$ |
| Te-3     | electron | 3    | -0.3006  | $1.466869 \times 10^9$     | $1.465476 \times 10^9$      | $9.5043 \times 10^{-4}$  |
| Te+3     | positron | 3    | -0.3006  | $1.466869 \times 10^9$     | $1.465476 \times 10^9$      | $9.5043 \times 10^{-4}$  |
| Te-4     | electron | 4    | 0.0000   | $2.419077 \times 10^9$     | $2.422187 \times 10^9$      | $-1.2839 \times 10^{-3}$ |
| Te+4     | positron | 4    | 0.0000   | $2.419077 \times 10^9$     | $2.422187 \times 10^9$      | $-1.2839 \times 10^{-3}$ |
| Te-5     | electron | 5    | 0.3006   | $1.472718 \times 10^9$     | $1.466912 \times 10^9$      | $3.9576 \times 10^{-3}$  |
| Te+5     | positron | 5    | 0.3006   | $1.472718 \times 10^9$     | $1.466912 \times 10^9$      | $3.9576 \times 10^{-3}$  |
| Te-6     | electron | 6    | 0.5994   | $3.271389 \times 10^8$     | $3.294436 \times 10^8$      | $-6.9957 \times 10^{-3}$ |
| Te+6     | positron | 6    | 0.5994   | $3.271389 \times 10^8$     | $3.294436 \times 10^8$      | $-6.9957 \times 10^{-3}$ |

$L_2$  relative error on  $|E|$ :  $4.769359 \times 10^{-3}$ ; absolute  $L_2$ :  $6.274261 \times 10^6$  V/m.